KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3331 | 3 Credits: 3 | Engineering Science | Source-grounded

# Analog & Digital Electronics

 To introduce various semiconductor diodes and applicationcircuits.

OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3331 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 3331.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

## Official course information

**Course information**

| Item | Official information |
|---|---|
| Programme | Diploma in Automation and Robotics |
| Course code | 3331 |
| Course title | Analog & Digital Electronics |
| Semester | 3 Credits: 3 |
| Course category | Engineering Science |
| Periods per week | 3 (L:3, T:0, P:0) Periods per semester: 45 |
| Periods per semester | 45 |

| Item | Official information |
| --- | --- |
| Credits | 3 |
| Prerequisite | See the official syllabus PDF |
| Assessment | Use the official assessment section reproduced below |

### Modules

4 official module sections detected.

### Topic entries

17 source-aligned topic entries retained.

### Study mode

Theory and application emphasis.

STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

### Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

# Objectives, outcomes and mapping

## Course objectives

-  To introduce various semiconductor diodes and applicationcircuits.
-  To illustrate the working of differenttransistors.
-  To introduce various linear and non linear wave shapingcircuits.

**Course outcomes**

| CO | Official outcome | Study level |
|---|---|---|
| CO1 | 10 Understanding Junction Transistor and transistor amplifier. Summarize the working principles of JFET, | See official syllabus |
| CO2 | 11 Understanding MOSFET and UJT. Design and implement Combinational and | See official syllabus |
| CO3 | Sequential logic circuits. 11 Applying Describe the process of Analog to Digital conversion | See official syllabus |
| CO4 | and Digital to Analog conversion. 11 Applying | See official syllabus |

**CO–PO mapping evidence**

| Row | Extracted official mapping |
|---|---|
| CO1 | CO1 2 |
| CO2 | CO2 2 |
| CO3 | CO3 2 |
| CO4 | CO4 3 2 |

COVERAGE AUDIT

# Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

**Module coverage**

| Module | Official heading | Topic entries | Hours |
|---|---|---|---|
| Module 1 | transistor amplifier. | 5 | As specified |
| Module 2 | Summarize the working principle of JFET, MOSFET and UJT | 4 | As specified |
| Module 3 | Design and implement combinational and sequential circuits. | 5 | As specified |
| Module 4 | Describe the process of analog to digital and digital analog conversion | 3 | As specified |

MODULE 1

# transistor amplifier.

This module is presented from the official syllabus scope for Analog & Digital Electronics. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: transistor amplifier.
- Official duration: As specified

- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

## Describe the physical structure of BJT

Describe the physical structure of BJT is an official topic in Module 1 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Describe the physical structure of BJT using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, describe the physical structure of bjt should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Describe the physical structure of BJT means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-[illegible] Describe the physical structure of BJT [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations

Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations is an official topic in Module 1 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the working principle of bjt. 2 understanding summarize various transistor configurations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## 2 Understanding and gain relations. Illustrate the characteristics ofBJT

2 Understanding and gain relations. Illustrate the characteristics ofBJT is an official topic in Module 1 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding and gain relations. Illustrate the characteristics ofBJT using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding and gain relations. illustrate the characteristics ofbjt should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 2 Understanding and gain relations. Illustrate the characteristics ofBJT means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 1-□□ 2 Understanding and gain relations. Illustrate the characteristics ofBJT □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

## 3 Understanding Explain transistor action and amplification

3 Understanding Explain transistor action and amplification is an official topic in Module 1 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding Explain transistor action and amplification using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 understanding explain transistor action and amplification should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Understanding Explain transistor action and amplification means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 1-□□ 3 Understanding Explain transistor action and amplification □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

**1 Understanding Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased transistor Transistor biasing - modes of operation - active, saturation and cut off - operation of NPN transistor - transistor currents Transistor configurations - current gain □□□ and □ - relationship between □ and □ - effect of temperature in leakagecurrent Input and output characteristics of common base and common emitter configuration - Cut off, Active and Saturation Regions - comparison of three configurations - importance of CEconfiguration Transistor action - Transistor as amplifier.**

1 Understanding Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased transistor Transistor biasing - modes of operation - active, saturation and cut off - operation of NPN transistor - transistor currents Transistor configurations - current gain □□□ and □ - relationship between □ and □ - effect of temperature in leakagecurrent Input and output characteristics of common base and common emitter configuration - Cut off, Active and Saturation Regions - comparison of three configurations - importance of CEconfiguration Transistor action - Transistor as amplifier. is an official topic in Module 1 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased transistor Transistor biasing - modes of operation - active, saturation and cut off - operation of NPN transistor - transistor currents Transistor configurations - current gain □□□ and □ - relationship between □ and □ - effect of temperature in leakagecurrent Input and output characteristics of common base and common emitter configuration - Cut off, Active and Saturation Regions - comparison of three configurations - importance of CEconfiguration Transistor action - Transistor as amplifier. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding bipolar junction transistors - structure - npn and pnp - symbol - unbiased transistor transistor biasing - modes of operation - active, saturation and cut off - operation of npn transistor - transistor currents transistor configurations - current gain □□□ and □ - relationship between □ and □ - effect of temperature in leakagecurrent input and output characteristics of common base and common emitter configuration - cut off, active and saturation regions - comparison of three configurations - importance of ceconfiguration transistor action - transistor as amplifier. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 1 Understanding Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased transistor Transistor biasing - modes of operation - active, saturation and cut off - operation of NPN transistor - transistor currents Transistor configurations - current gain □□□ and □ - relationship between □ and □ - effect of temperature in leakagecurrent Input and output characteristics of common base and common emitter configuration - Cut off, Active and Saturation Regions - comparison of three configurations - importance of CEconfiguration Transistor action - Transistor as amplifier. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 1 Understanding Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased transistor Transistor biasing - modes of operation - active, saturation and cut off - operation of NPN transistor - transistor currents Transistor configurations - current gain □□□ and □ - relationship between □ and □ - effect of temperature in leakagecurrent Input and output characteristics of common base and common emitter configuration - Cut off, Active and Saturation Regions - comparison of three configurations - importance of CEconfiguration Transistor action - Transistor as amplifier. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
transistor amplifier.

   M1.01      Describe the physical structure of BJT                2
Understanding

   M1.02      Explain the working principle of BJT.                 2
Understanding

              Summarize various transistor configurations
   M1.03                                                            2
Understanding
              and gain relations.
              Illustrate the characteristics ofBJT
   M1.04                                                            3
Understanding

              Explain transistor action and amplification
   M1.05                                                            1
Understanding

 Contents:
 Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased
transistor
 Transistor biasing - modes of operation - active, saturation and cut off -
operation of NPN
 transistor - transistor currents
 Transistor configurations - current gain □□□ and □ - relationship between □ and
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

## Summarize the working principle of JFET, MOSFET and UJT

This module is presented from the official syllabus scope for Analog & Digital Electronics. Use the topic cards for learning and the source transcript for exact

coverage.

**As specified**
official hours

**CO-linked**
see outcomes

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## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Summarize the working principle of JFET, MOSFET and UJT
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## 3 Understanding MOSFET and UJT Explain the working principle of JFET,

3 Understanding MOSFET and UJT Explain the working principle of JFET, is an official topic in Module 2 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding MOSFET and UJT Explain the working principle of JFET, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 understanding mosfet and ujt explain the working principle of jfet, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Understanding MOSFET and UJT Explain the working principle of JFET, means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 2-□□ 3 Understanding MOSFET and UJT Explain the working principle of JFET, □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## 3 Understanding MOSFET and UJT Illustrate the characteristics of JFET,

3 Understanding MOSFET and UJT Illustrate the characteristics of JFET, is an official topic in Module 2 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding MOSFET and UJT Illustrate the characteristics of JFET, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 understanding mosfet and ujt illustrate the characteristics of jfet, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Understanding MOSFET and UJT Illustrate the characteristics of JFET, means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 2-[illegible] 3 Understanding MOSFET and UJT Illustrate the characteristics of JFET, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## 4 Understanding MOSFET and UJT

4 Understanding MOSFET and UJT is an official topic in Module 2 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding MOSFET and UJT using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding mosfet and ujt should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 4 Understanding MOSFET and UJT means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

> **Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-[illegible] 4 Understanding MOSFET and UJT [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

**Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator**

Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator is an official topic in Module 2 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, compare bjt and fet 1 remembering junction field effect transistors: structure - symbol - operation of n channel jfet - drain characteristics - transfer characteristics - comparison with bjt. mosfet: structure - symbol - principle of operation (n channel depletion type only), drain characteristics - transfer characteristics. ujt: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - vi characteristics. ujt relaxation oscillator should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize the working principle of JFET, MOSFET and UJT
             Describe the physical structure of JFET,
  M2.01                                                               3
Understanding
             MOSFET and UJT
             Explain the working principle of JFET,
  M2.02                                                               3
Understanding
             MOSFET and UJT
             Illustrate the characteristics of JFET,
  M2.03                                                               4
Understanding
             MOSFET and UJT

  M2.04      Compare BJT and FET                                      1
Remembering

             Series Test 1                                            1

Contents:
Junction Field Effect Transistors: Structure - symbol - operation of N channel
JFET - drain
characteristics - transfer characteristics - comparison with BJT.
MOSFET: Structure - symbol - principle of operation (N channel depletion type
only),
drain characteristics - transfer characteristics.
UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio -
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

# Design and implement combinational and sequential circuits.

This module is presented from the official syllabus scope for Analog & Digital Electronics. Use the topic cards for learning and the source transcript for exact

coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Design and implement combinational and sequential circuits.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

## Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer,

Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer, is an official topic in Module 3 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, simplification of logic function using k-map 1 remembering explain the working of multiplexer, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer, means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## 2 Understanding demultiplexer, adder and subtractor.

2 Understanding demultiplexer, adder and subtractor. is an official topic in Module 3 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding demultiplexer, adder and subtractor. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding demultiplexer, adder and subtractor. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 2 Understanding demultiplexer, adder and subtractor. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] 2 Understanding demultiplexer, adder and subtractor. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

### Develop BCD adder and Carry look head 2 Applying adder

Develop BCD adder and Carry look head 2 Applying adder is an official topic in Module 3 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify Develop BCD adder and Carry look head 2 Applying adder using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, develop bcd adder and carry look head 2 applying adder should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Develop BCD adder and Carry look head 2 Applying adder means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3-□□ Develop BCD adder and Carry look head 2 Applying adder □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## Understand the working of different Latches 3 Applying and Flipflops

Understand the working of different Latches 3 Applying and Flipflops is an official topic in Module 3 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Understand the working of different Latches 3 Applying and Flipflops using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, understand the working of different latches 3 applying and flipflops should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Understand the working of different Latches 3 Applying and Flipflops means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3-□□ Understand the working of different Latches 3 Applying and Flipflops □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

**Design shift registers, synchronous and 3 Applying asynchronous counters using flipflops simplification of logic functions using K-map, minimization of logical functions.Don't care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder. 1 bit memory, Clocked SR JK D flipflops,Shift registers, Synchronous and asynchronous counters. Sequence generator.**

Design shift registers, synchronous and 3 Applying asynchronous counters using flipflops simplification of logic functions using K-map, minimization of logical functions.Don't care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder. 1 bit memory, Clocked SR JK D flipflops,Shift registers, Synchronous and asynchronous counters. Sequence generator. is an official topic in Module 3 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Design shift registers, synchronous and 3 Applying asynchronous counters using flipflops simplification of logic functions using K-map, minimization of logical functions.Don't care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder. 1 bit memory, Clocked SR JK D flipflops,Shift registers, Synchronous and asynchronous counters. Sequence generator. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, design shift registers, synchronous and 3 applying asynchronous counters using flipflops simplification of logic functions using k-map, minimization of logical functions.don't care conditions, multiplexer, de-multiplexer/decoders, adders, subtractors, bcd arithmetic, carry look ahead adder. 1 bit memory, clocked sr jk d flipflops,shift registers, synchronous and asynchronous counters. sequence generator. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Design shift registers, synchronous and 3 Applying asynchronous counters using flipflops simplification of logic functions using K-map, minimization of logical functions.Don’t care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder. 1 bit memory, Clocked SR JK D flipflops,Shift registers, Synchronous and asynchronous counters. Sequence generator. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ Design shift registers, synchronous and 3 Applying asynchronous counters using flipflops simplification of logic functions using K-map, minimization of logical functions.Don’t care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder. 1 bit memory, Clocked SR JK D flipflops,Shift registers, Synchronous and asynchronous counters. Sequence generator. □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

**Official syllabus detail and terminology**

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Design and implement combinational and sequential circuits.

  M3.01      Simplification of logic function using K-map            1
Remembering

             Explain the working of multiplexer,
  M3.02                                                              2
Understanding
             demultiplexer, adder and subtractor.

  M3.03      Develop BCD adder and Carry look head                   2
Applying
             adder

  M3.04     Understand the working of different Latches              3
Applying
            and Flipflops

  M3.05      Design shift registers, synchronous and                 3
Applying
             asynchronous counters using flipflops
Contents:

simplification of logic functions using K-map, minimization of logical
functions.Don’t care conditions, Multiplexer, De-Multiplexer/Decoders, Adders,
Subtractors, BCD arithmetic, carry look ahead adder.
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

# Describe the process of analog to digital and digital analog conversion

This module is presented from the official syllabus scope for Analog & Digital Electronics. Use the topic cards for learning and the source transcript for exact

coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Describe the process of analog to digital and digital analog conversion
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## 3 Understanding conversion. Explain the working of various digital

3 Understanding conversion. Explain the working of various digital is an official topic in Module 4 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding conversion. Explain the working of various digital using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 understanding conversion. explain the working of various digital should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Understanding conversion. Explain the working of various digital means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 4-□□ 3 Understanding conversion. Explain the working of various digital □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## 4 Understanding to analog convertors Explain the working of various analog

4 Understanding to analog convertors Explain the working of various analog is an official topic in Module 4 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding to analog convertors Explain the working of various analog using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding to analog convertors explain the working of various analog should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 4 Understanding to analog convertors Explain the working of various analog means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 4-□□ 4 Understanding to analog convertors Explain the working of various analog □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

**4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 https://www.electronics-tutorials.ws 2 https://www.elprocus.com 3 http://www.brainkart.com/menu/electronics-engineering/ 4 https://www.electrical4u.com 5 https://www. tutorialspoint.com**

4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 https://www.electronics-tutorials.ws 2 https://www.elprocus.com 3 http://www.brainkart.com/menu/electronics-engineering/ 4 https://www.electrical4u.com 5 https://www. tutorialspoint.com is an official topic in Module 4 of Analog & Digital Electronics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied

Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 https://www.electronics-tutorials.ws 2 https://www.elprocus.com 3 http://www.brainkart.com/menu/electronics-engineering/ 4 https://www.electrical4u.com 5 https://www. tutorialspoint.com using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 4 understanding to digital convertors analog to digital conversion: sampling, quantization, encoding digital to analog convertors: weighted resistor, r-2r ladder analog to digital convertors: counter type, successive approximation, dual slope, example of a to d and d to a convertor ics text /reference: t/r booktitle/author t1 r s sedha - a text book of applied electronics - s chand n n bhargava, kulshreshtha and s c gupta - basic electronics and linear r1 circuits- tmh r2 robert boylestad - electronic devices and circuits - phi anand kumar- fundamentals of digital circuits r3 r4 david a bell - electronic devices and circuits - phi online resources: sl. no website link 1 https://www.electronics-tutorials.ws 2 https://www.elprocus.com 3 http://www.brainkart.com/menu/electronics-engineering/ 4 https://www.electrical4u.com 5 https://www. tutorialspoint.com should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 https://www.electronics-tutorials.ws 2 https://www.elprocus.com 3 http://www.brainkart.com/menu/electronics-engineering/ 4 https://www.electrical4u.com 5 https://www. tutorialspoint.com means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ 4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 https://www.electronics-tutorials.ws 2 https://www.elprocus.com 3 http://www.brainkart.com/menu/electronics-engineering/ 4 https://www.electrical4u.com 5 https://www. tutorialspoint.com □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

**Official syllabus detail and terminology**

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Describe the process of analog to digital and digital analog conversion

               Describe the process of analog to digital
   M4.01                                                                     3
Understanding
               conversion.
                 Explain the working of various digital
    M4.02                                                                4
Understanding
                 to analog convertors
                 Explain the working of various analog
    M4.03                                                                4
Understanding
                 to digital convertors

                 Series Test 2                                           1

  Contents:
  Analog to Digital Conversion: Sampling, Quantization, Encoding
  Digital to Analog Convertors: Weighted resistor, R-2R ladder
  Analog to Digital Convertors: Counter type, Successive approximation, Dual
slope,
  Example of A to D and D to A convertor ICs

Text /Reference:

    T/R                                        BookTitle/Author
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

## Concept and formula bank

### Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

## Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

## Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

## Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

**Define or explain Describe the physical structure of BJT. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Describe the physical structure of BJT*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain the working principle of BJT. 2 Understanding Summarize various transistor configurations*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 2 Understanding and gain relations. Illustrate the characteristics ofBJT. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding and gain relations. Illustrate the characteristics ofBJT*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 3 Understanding Explain transistor action and amplification. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding Explain transistor action and amplification*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 2

**Define or explain 3 Understanding MOSFET and UJT Explain the working principle of JFET,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding MOSFET and UJT Explain the working principle of JFET,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 3 Understanding MOSFET and UJT Illustrate the characteristics of JFET,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding MOSFET and UJT Illustrate the characteristics of JFET,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 4 Understanding MOSFET and UJT. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding MOSFET and UJT*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Compare BJT and FET 1 Remembering Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT. MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics. UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 3

**Define or explain Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 2 Understanding demultiplexer, adder and subtractor.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding demultiplexer, adder and subtractor.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Develop BCD adder and Carry look head 2 Applying adder. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Develop BCD adder and Carry look head 2 Applying adder*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Understand the working of different Latches 3 Applying and Flipflops. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Understand the working of different Latches 3 Applying and Flipflops*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 4

**Define or explain 3 Understanding conversion. Explain the working of various**

**digital. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding conversion. Explain the working of various digital*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 4 Understanding to analog convertors Explain the working of various analog. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding to analog convertors Explain the working of various analog*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 https://www.electronics-tutorials.ws 2 https://www.elprocus.com 3 http://www.brainkart.com/menu/electronics-engineering/ 4 https://www.electrical4u.com 5 https://www. tutorialspoint.com. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding to digital convertors Analog to Digital Conversion: Sampling, Quantization, Encoding Digital to Analog Convertors: Weighted resistor, R-2R ladder Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs*

*Text /Reference: T/R BookTitle/Author T1 R S Sedha - A Text Book of Applied Electronics - S Chand N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear R1 Circuits- TMH R2 Robert Boylestad - Electronic Devices and Circuits - PHI Anand Kumar- Fundamentals of Digital Circuits R3 R4 David A Bell - Electronic Devices and Circuits - PHI Online resources: Sl. No Website Link 1 https://www.electronics-tutorials.ws 2 https://www.elprocus.com 3 http://www.brainkart.com/menu/electronics-engineering/ 4 https://www.electrical4u.com 5 https://www. tutorialspoint.com*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

## Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **Describe the physical structure of BJT**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **3 Understanding MOSFET and UJT Explain the working principle of JFET,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **Simplification of logic function using K-map 1 Remembering Explain the working of multiplexer,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 4

1-mark definition: State the meaning of **3 Understanding conversion. Explain the working of various digital**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

## Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.

- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

# Glossary

**Study glossary**

| Term | Meaning in this lesson |
| --- | --- |
| Course Outcome | A capability the student should demonstrate after completing the course. |
| Module | An official syllabus unit with defined topics and hours. |
| CIE | Continuous Internal Evaluation. |
| ESE | End Semester Examination. |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

SOURCE-SUPPORTED REFERENCES

# References and web resources

## References listed in the official PDF

References are included in the official source PDF.

## Web resources listed in the official PDF

- Sl. No Website Link https://www.electronics-tutorials.ws https://www.elprocus.com
- http://www.brainkart.com/menu/electronics-engineering/ https://www.electrical4u.com https://www. tutorialspoint.com

IN-PAGE SEARCH

# Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3331. Verify institutional updates against the current official curriculum.